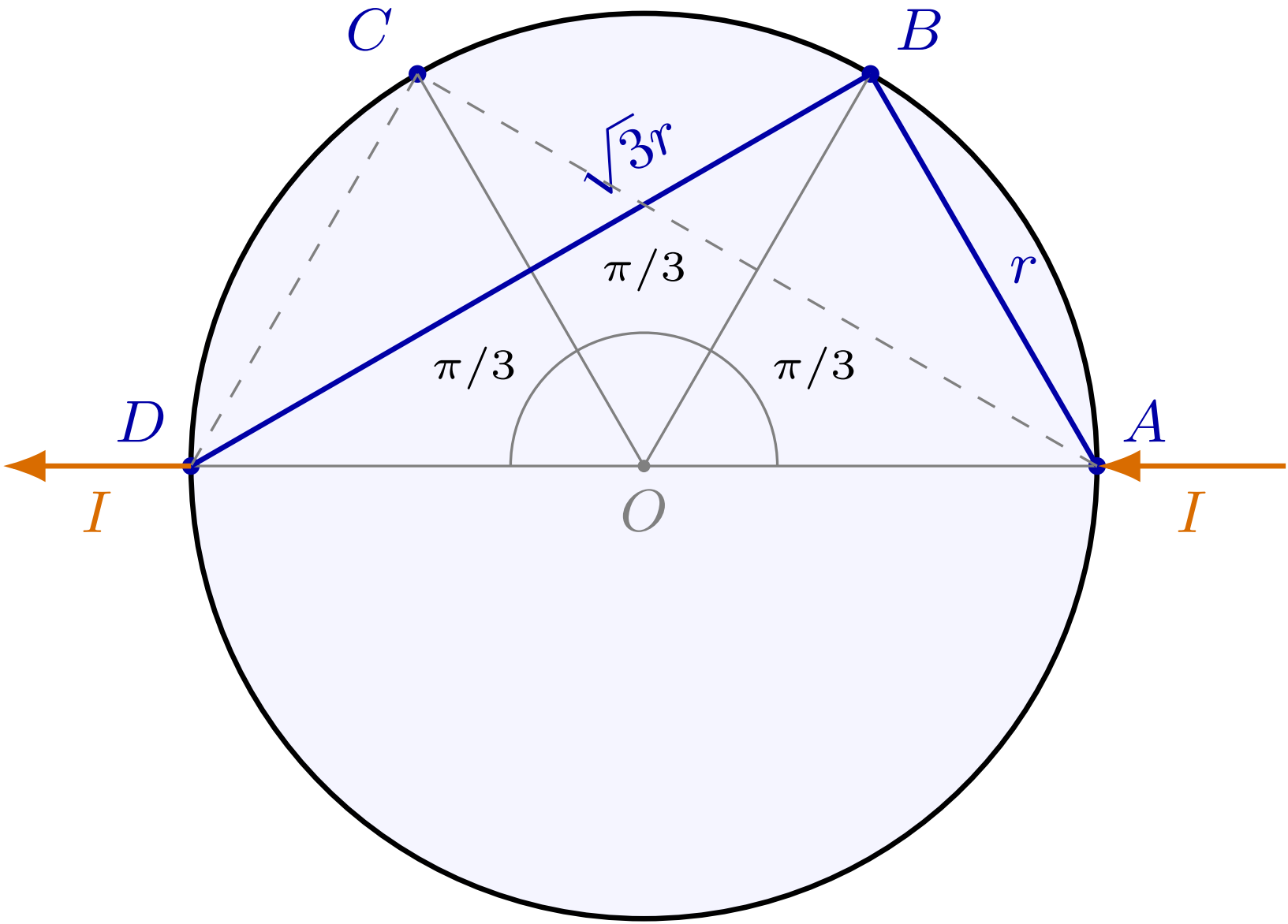

$$U = V_B - V_C$$